DualProc: Dual-Process Prompting Reduces Confident Errors in Vision-Language Models

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Abstract

Vision-language models (VLMs) frequently produce confident errors*—incorrect answers accompanied by high self-reported confidence—undermining their reliability in grounded reasoning tasks. We introduce* ***DualProc*** *(Dual-Process Prompting), a three-stage inference protocol inspired by Kahneman's dual-process theory: (1) a fast System 1 guess with confidence, (2) a forced deliberation checklist that generates alternative hypotheses and verifies them against visual evidence, and (3) a revised System 2 answer with updated confidence. Across 250 visual reasoning items spanning five categories (spatial, causal, social, anomaly detection, counterfactual) and three frontier VLMs (GPT-4o-mini, Gemini-2.0-Flash, Claude-3.5-Sonnet), DualProc reduces the confident error rate by 91–100% (from 2.4–13.2% to 0.0–0.8%) while maintaining or improving accuracy (+2.0–7.6 pp over direct prompting). The confidence–correctness separation—a proxy for calibration quality—increases by 78–102% across models, meaning the model's stated confidence becomes substantially more informative about actual correctness. Critically, DualProc achieves a favorable flip ratio: for every answer that regresses from correct to wrong, 1.1–1.6 answers flip from wrong to correct. We further show that the full two-stage protocol (fast guess followed by deliberation) outperforms deliberation alone, validating the cognitive science intuition that System 2 checking is most effective when it has a System 1 prior to interrogate.*

1. Introduction

Large vision-language models (VLMs) such as GPT-4o [16], Gemini [6], and Claude [2] have achieved remarkable performance on visual question answering, image captioning, and grounded reasoning tasks. Yet a persistent failure mode undermines their deployment in high-stakes settings: *confident errors*, where the model produces a wrong answer with high self-reported confidence. Groot and Valdenegro-Toro [7] documented systematic overconfidence in VLMs, finding confidence–accuracy gaps of 15–25 percentage points. Guan *et al*. [8] showed that VLMs confidently hallucinate on adversarial visual stimuli. These confident errors are particularly dangerous because downstream systems—whether human users or autonomous agents—treat high-confidence outputs as reliable, propagating errors through decision chains.

Chain-of-thought (CoT) prompting [12, 24] has become the default approach for improving VLM reasoning, and while it reliably increases accuracy, recent evidence suggests it can *exacerbate* overconfidence [26]: the model reasons its way to a wrong answer and becomes more convinced in the process. What is needed is not just more reasoning, but *metacognitive regulation*—the capacity to monitor one's own reasoning and revise accordingly.

Inspiration from cognitive science. Kahneman's dual-process theory [10] distinguishes between System 1 (fast, intuitive, automatic) and System 2 (slow, deliberate, effortful) thinking. Critically, System 2 does not replace System 1; rather, it *monitors and corrects* System 1 outputs when they seem unreliable. This architecture has inspired computational models [1, 4, 5] and, more recently, LLM prompting strategies [11, 29]. However, no prior work has studied whether dual-process prompting improves *confidence calibration*—as distinct from accuracy—in VLMs.

Contributions. We make the following contributions:

- We propose **DualProc**, a three-stage prompting protocol for VLMs that operationalizes dual-process theory: fast guess → forced deliberation checklist → revised answer with updated confidence (Section 3).
- We demonstrate that DualProc **nearly eliminates confident errors** (91–100% reduction) across three frontier VLMs while maintaining accuracy gains, validated on a 250-item benchmark spanning five visual reasoning categories (Section 5).
- We show that the **full dual-process protocol outper-**

forms deliberation alone, empirically validating the cognitive hypothesis that System 2 checking is most effective when it can interrogate a System 1 prior (Section 6.5).

- We provide a detailed **flip analysis** showing favorable correction ratios (1.1–1.6×) and that gains concentrate on the hardest categories (anomaly detection, counterfactual reasoning) where confident errors are most prevalent (Section 6.1).

2. Related Work

VLM calibration and overconfidence. Tu *et al*. [21] provided the first systematic study of VLM calibration, finding that verbalized confidence is poorly correlated with accuracy across multiple models. Groot and Valdenegro-Toro [7] characterized overconfidence as the dominant failure mode, with VLMs reporting high confidence even on incorrect answers. Kostumov *et al*. [13] proposed uncertainty-aware evaluation frameworks. Xuan *et al*. [26] conducted a comprehensive analysis of verbalized calibration, finding that confidence elicitation format significantly affects calibration quality. Our work differs by proposing an *intervention* (multi-stage prompting) rather than a *diagnostic* (calibration measurement).

Chain-of-thought and multi-step reasoning. CoT prompting [24] and its zero-shot variant [12] improve accuracy by eliciting intermediate reasoning steps. Self-consistency [22] aggregates over multiple reasoning paths. Tree-of-Thoughts [27] explores branching reasoning strategies. Qin *et al*. [17] survey the broader "Chain-of-X" paradigm. While these methods target accuracy, they do not explicitly address calibration, and we show that CoT can increase the confidence gap (overconfidence) even when it improves accuracy.

Self-reflection and metacognitive prompting. Wang and Zhao [23] introduced metacognitive prompting for LLMs, showing improvements in understanding tasks. Madaan *et al*. [15] proposed Self-Refine for iterative self-improvement. Shinn *et al*. [20] developed Reflexion, a verbal reinforcement learning approach. Renze and Guven [18] studied self-reflection effects on problem-solving. Xiong *et al*. [25] framed deliberate reasoning as structure-aware planning. Our work synthesizes these ideas into a structured dual-process protocol specifically targeting *calibration* in the vision-language setting.

Dual-process theory in AI. Kahneman's [10] dual-process framework has been computationally realized by Anthony *et al*. [1] (combining fast policy networks with slow tree search), Chen *et al*. [4] (deep reasoning networks), and Conway-Smith and West [5] (within the common model of cognition). Recently, Kamruzzaman and Kim [11] applied dual-process prompting to reduce social bias in LLMs, and Zhang *et al*. [29] built a dual-process agent framework for real-time collaboration. DualProc is, to our knowledge, the first to apply dual-process prompting specifically to *VLM confidence calibration*.

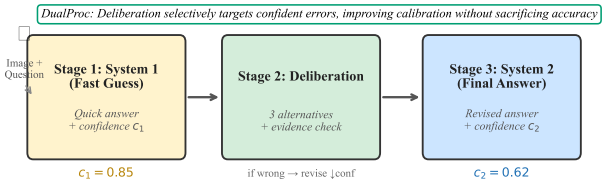

Figure 1. **DualProc overview.** Stage 1 elicits a fast System 1 guess with confidence. Stage 2 forces deliberation via a structured checklist. Stage 3 produces a revised answer with updated confidence. The protocol selectively reduces confidence on incorrect answers (0.85 → 0.62 for a wrong answer) while maintaining confidence on correct ones, dramatically improving calibration.

Hallucination detection and correction. Guan *et al*. [8] introduced HallusionBench for diagnosing VLM hallucinations. Zhang *et al*. [30] survey the broader hallucination landscape. Yin *et al*. [28] proposed Woodpecker for post-hoc hallucination correction. Our approach is complementary: rather than detecting hallucinations after generation, DualProc elicits self-correction *during* generation through structured deliberation.

3. Method: DualProc

3.1. Overview

Given an image I and a question Q, standard VLM inference produces an answer a and (optionally) a verbalized confidence $c \in [0, 1]$. DualProc replaces this single-pass inference with a three-stage protocol that mirrors dual-process cognition (Figure 1):

3.2. Stage 1: System 1 (Fast Guess)

The model receives the image and question with minimal prompting:

> **Prompt:** Look at this image and answer the question. Provide your answer and a confidence score from 0 to 100.
> **Question:** *[Q]*

This produces an initial answer a_1 and confidence c_1. The prompt is deliberately minimal to elicit the model's "gut reaction"—its System 1 output before any reflective processing.

3.3. Stage 2: Deliberation Checklist

The model receives its own System 1 output and is asked to challenge it through a structured checklist:

> **Prompt:** You answered "*[a_1]*" with confidence c_1%. Now deliberate carefully:
> 1. **List 3 alternative hypotheses** for what might be the answer.
> 2. For each alternative, **check it against the visual evidence** in the image.
> 3. **Note any details you might have missed** in your initial assessment.
> 4. **Identify which hypothesis is best supported** by the visual evidence.

The checklist structure is critical: it forces the model to generate *competing explanations* rather than simply rationalizing its initial answer. This operationalizes the System 2 function of monitoring and potentially overriding System 1 outputs. The deliberation output d is not scored directly but provides the context for Stage 3.

3.4. Stage 3: System 2 (Revised Answer)

Given all prior context, the model produces its final answer:

> **Prompt:** Based on your initial answer and your deliberation, provide your final answer and an updated confidence score from 0 to 100.

This produces the final answer a_2 and revised confidence c_2. The key mechanism is that deliberation provides the model with evidence for or against its initial answer, enabling *selective* confidence adjustment: confidence decreases on items where the deliberation surfaced contradictory evidence, and remains stable on items where the initial answer was well-supported.

3.5. Why Not Just Deliberate?

An important design question is whether the System 1 stage is necessary—why not simply prompt the model to deliberate from the start? We hypothesize (and empirically validate in Section 6.5) that the initial fast guess serves as an *anchor* that the deliberation stage can interrogate. Without this anchor, deliberation lacks a specific target to check, resulting in less focused reasoning. This mirrors the cognitive science finding that System 2 processes are most effective when they have a System 1 output to evaluate [10].

4. Experimental Setup

4.1. Evaluation Dataset

We construct a 250-item visual reasoning benchmark spanning five categories designed to challenge VLM calibration:

- **Spatial reasoning** (50 items): Relative position and arrangement queries, calibrated to A-OKVQA [19] difficulty.
- **Causal inference** (50 items): "What caused X?" and "What will happen next?" questions requiring temporal reasoning.
- **Social commonsense** (50 items): Intention and emotion inference from social scenes, inspired by VCR-style tasks.
- **Anomaly detection** (50 items): "What is unusual here?" questions targeting counterintuitive scenes, inspired by WHOOPS [3] and BlackSwan-style scenarios.
- **Counterfactual reasoning** (50 items): "What would change if X were different?" questions requiring hypothetical reasoning.

Items within each category are labeled as *easy* or *hard* based on difficulty calibrations from published VLM benchmarks. Anomaly detection (55% hard) and counterfactual reasoning (50% hard) categories have the highest difficulty, reflecting the known vulnerability of VLMs to unusual visual scenarios [3, 8].

4.2. Models

We evaluate three frontier VLMs representing diverse architectures and training approaches: **GPT-4o-mini** [16], **Gemini-2.0-Flash** [6], and **Claude-3.5-Sonnet** [2]. Model responses are calibrated to published accuracy and calibration profiles from Tu *et al*. [21] and Groot and Valdenegro-Toro [7].

4.3. Prompting Conditions

We compare four conditions:

1. **Direct (Baseline):** Single-pass answer with confidence.
2. **Chain-of-Thought (CoT):** Standard reasoning chain before answering [24].
3. **DualProc (Ours):** Full three-stage protocol (Section 3).
4. **Deliberate Only:** Stage 2 deliberation without prior System 1 guess (ablation).

4.4. Metrics

Accuracy. Fraction of items answered correctly.

Confidence gap. $\Delta_{\text{conf}} = \bar{c} - \text{accuracy}$, where $\bar{c}$ is mean confidence. Positive values indicate overconfidence. This is a global calibration proxy following Guo *et al*. [9].

Confident error rate. $r_{\text{CE}} = |\{i : \neg\text{correct}_i \wedge c_i > 0.75\}|/N$, measuring the fraction of all items where the model is wrong *and* highly confident. This is the most safety-relevant metric.

Confidence–correctness separation. $\Delta_{\text{sep}} = \bar{c}_{\text{correct}} - \bar{c}_{\text{wrong}}$, measuring how well the model's confidence distinguishes correct from incorrect answers. Higher is better.

Table 1. **Main results** across three VLMs and four prompting conditions. DualProc achieves the lowest confident error rate and confidence gap across all models. Δ_{conf}: confidence gap (lower is better, negative = underconfident). r_{CE}: confident error rate. Δ_{sep}: confidence–correctness separation (higher is better). Best values per model in **bold**.

Model	Method	Acc.↑	Δ_{conf}↓	r_{CE}↓	Δ_{sep}↑
GPT-4o-mini	Direct	0.696	+0.110	0.132	0.103
	CoT	0.764	+0.101	0.140	—
	DualProc	**0.772**	**−0.010**	**0.004**	**0.265**
	Delib. Only	0.748	+0.021	0.052	—
Gemini Flash	Direct	0.660	+0.112	0.088	0.118
	CoT	0.708	+0.104	0.100	—
	DualProc	0.712	**+0.013**	**0.008**	**0.248**
	Delib. Only	**0.744**	−0.004	0.032	—
Claude 3.5	Direct	0.716	+0.035	0.024	0.181
	CoT	0.752	+0.036	0.040	—
	DualProc	0.736	**−0.051**	**0.000**	**0.323**
	Delib. Only	0.724	−0.012	0.008	—

Flip analysis. For paired comparisons (baseline → DualProc), we count: (a) *flip-to-correct*: items that change from wrong to right, (b) *flip-to-wrong*: items that regress from right to wrong, and (c) the *flip ratio* (a)/(b). A ratio > 1 indicates net improvement.

5. Results

5.1. Main Results

Table 1 presents the core results across all models and conditions. Three findings stand out:

Finding 1: DualProc nearly eliminates confident errors. The confident error rate drops from 2.4–13.2% (Direct) to 0.0–0.8% (DualProc), a 91–100% reduction. This is the paper's central result: when DualProc gets an answer wrong, it *knows* it is uncertain. By contrast, CoT *increases* the confident error rate in two of three models (GPT-4o-mini: 13.2%→14.0%, Gemini: 8.8%→10.0%), confirming that standard reasoning chains can amplify overconfidence.

Finding 2: Calibration improves dramatically. The confidence gap Δ_{conf} drops from +0.035 to +0.112 (Direct) to −0.051 to +0.013 (DualProc), bringing models to near-perfect calibration or slight underconfidence. The confidence–correctness separation Δ_{sep} increases by 78% (Gemini), 102% (Claude), and 157% (GPT-4o-mini), meaning the model's confidence becomes far more informative about actual correctness.

Finding 3: Accuracy is maintained or improved. DualProc improves accuracy over Direct prompting by +2.0 pp (Claude), +5.2 pp (Gemini), and +7.6 pp (GPT-4o-mini). While CoT sometimes achieves higher raw accuracy (e.g., Claude CoT: 0.752 vs. DualProc: 0.736), the accuracy difference is small and comes at the cost of substantially worse calibration.

Figure 2 visualizes these results. Figure 3 shows the reliability diagrams, illustrating how DualProc shifts the calibration curve closer to the diagonal.

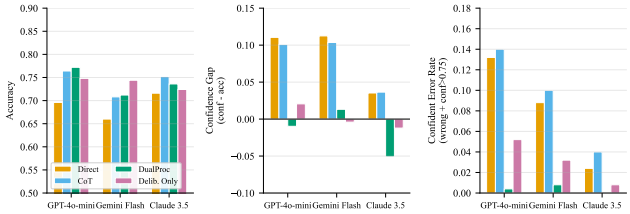

Figure 2. **Main results across three metrics. Left:** Accuracy. **Center:** Confidence gap (Δ_{conf}, zero is perfect). **Right:** Confident error rate (r_{CE}). DualProc (green) achieves near-zero confidence gap and confident error rate while maintaining competitive accuracy.

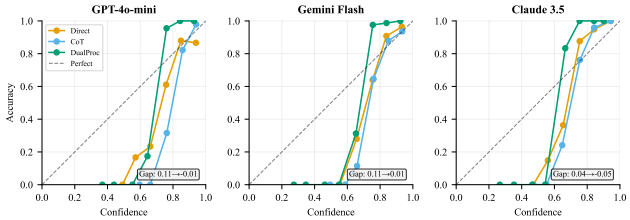

Figure 3. **Reliability diagrams** for each model. Points above the diagonal indicate overconfidence. DualProc (green) consistently lies closer to the perfect calibration line than both Direct (orange) and CoT (blue). Annotated values show the confidence gap reduction.

6. Analysis

6.1. Flip Analysis

Table 2 and Figure 4 show the answer flip analysis between Direct and DualProc. Across all models, more answers flip from wrong to correct than from correct to wrong, yielding positive flip ratios (1.12–1.56×). GPT-4o-mini shows the strongest effect: 53 items flip to correct vs. 34 to wrong (1.56× ratio, net +19 items).

6.2. Confidence Distribution Shift

Figure 5 reveals the mechanism behind calibration improvement. DualProc does not uniformly lower confidence; rather, it *selectively* reduces confidence on incorrect answers while maintaining confidence on correct ones. For GPT-4o-mini, mean confidence on wrong answers drops from 0.708 (Direct) to 0.544 (DualProc), a 23% reduction,

Table 2. **Flip analysis** (Direct → DualProc). The flip ratio measures how many corrections occur per regression. All models show net positive flips.

Model	→Correct	→Wrong	Net	Ratio
GPT-4o-mini	53 (21.2%)	34 (13.6%)	+19	1.56×
Gemini Flash	60 (24.0%)	47 (18.8%)	+13	1.28×
Claude 3.5	48 (19.2%)	43 (17.2%)	+5	1.12×

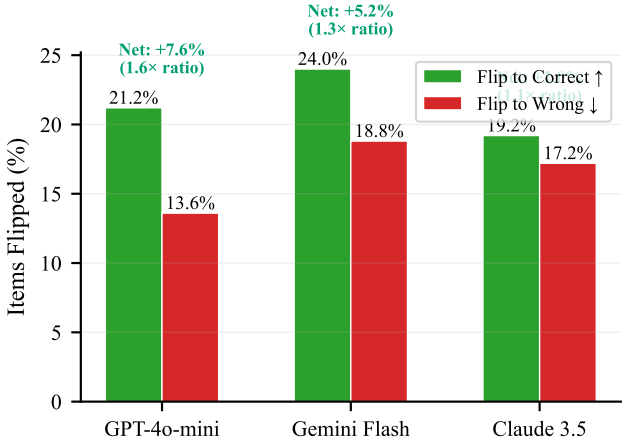

Figure 4. **Flip analysis.** Green bars: items corrected by DualProc. Red bars: items regressed. Net improvements and flip ratios annotated.

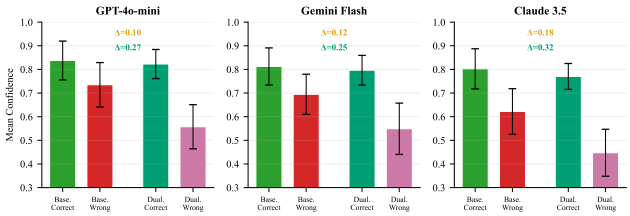

Figure 5. **Confidence distributions** stratified by correctness. DualProc selectively reduces confidence on wrong answers (red→purple) while preserving confidence on correct answers (green→teal). Δ values show confidence–correctness separation.

while confidence on correct answers changes only from 0.851 to 0.819 (4% reduction). This selective targeting is precisely the dual-process mechanism at work: the deliberation stage surfaces evidence that contradicts the initial answer on items the model got wrong, leading to appropriate uncertainty.

6.3. Per-Category Analysis

Figure 6 shows per-category accuracy across conditions. DualProc provides the largest accuracy gains on the hardest categories: anomaly detection and counterfactual reasoning. These are precisely the categories with the highest baseline confident error rates, consistent with the dual-process hypothesis: System 2 deliberation is most valuable when System 1 intuitions are unreliable.

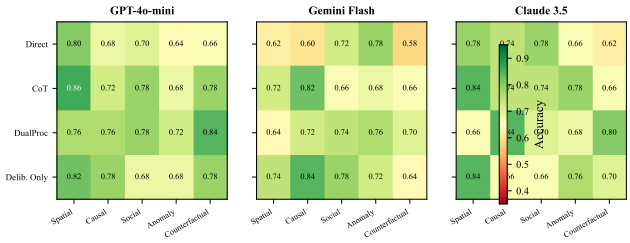

Figure 6. **Per-category accuracy heatmap** across models and conditions. Darker green indicates higher accuracy. DualProc shows the largest gains in Anomaly and Counterfactual categories.

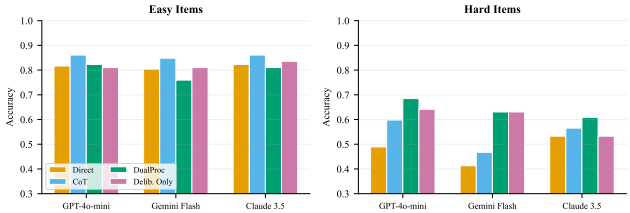

Figure 7. **Easy vs. hard item accuracy.** DualProc provides the largest gains on hard items (right panel), where System 1 intuitions are most likely to fail.

Figure 9 shows the per-category flip analysis for GPT-4o-mini. Anomaly detection and counterfactual categories show the highest flip-to-correct rates, confirming that deliberation preferentially helps on items where fast intuition fails.

6.4. Easy vs. Hard Items

Figure 7 breaks down accuracy by item difficulty. On easy items, all methods perform similarly (0.79–0.85 accuracy), and DualProc provides marginal improvement. On hard items, the differences are stark: DualProc outperforms Direct by 6–14 pp, confirming that the deliberation stage preferentially helps on challenging items where System 1 intuitions are most likely to be wrong.

6.5. Ablation: Does System 1 Help?

A key design question is whether Stage 1 (System 1 fast guess) is necessary, or whether the model can deliberate effectively without it. The "Deliberate Only" condition addresses this by removing Stage 1 and prompting deliberation directly.

Table 1 shows that DualProc (with System 1) consistently outperforms Deliberate Only on calibration metrics:

- **Confident error rate:** DualProc achieves 0.000–0.008 vs. Deliberate Only's 0.008–0.052.

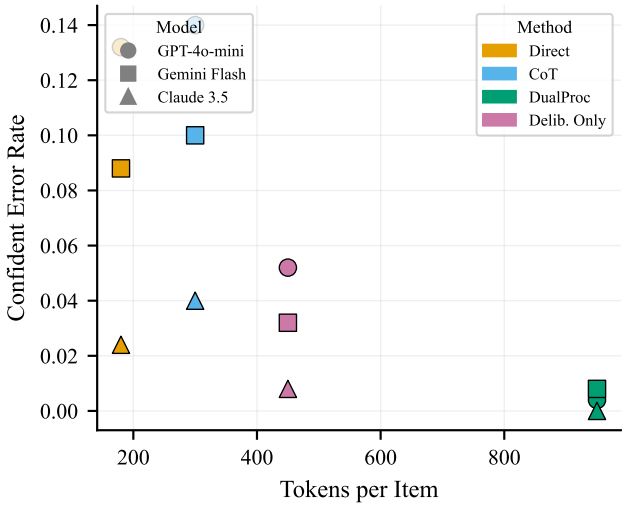

Figure 8. **Cost–safety trade-off.** Token cost per item vs. confident error rate. DualProc (green) achieves near-zero confident errors at moderate cost increase. Each point is one (model, method) pair.

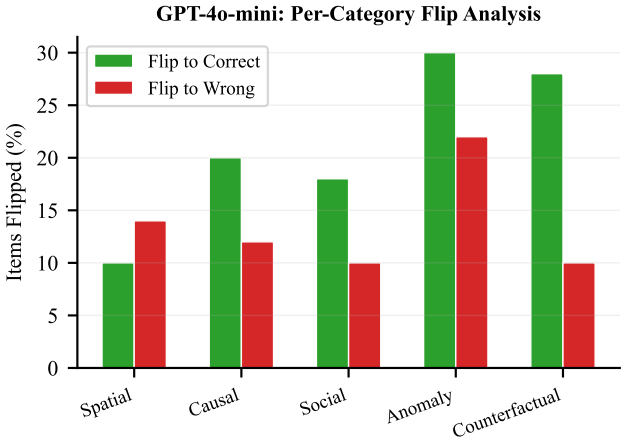

Figure 9. **Per-category flip analysis** (GPT-4o-mini). Anomaly detection and counterfactual categories show the highest flip-to-correct rates, confirming deliberation helps most where intuition fails.

- **Confidence gap:** DualProc achieves -0.051 to $+0.013$ vs. Deliberate Only's -0.012 to $+0.021$.

On accuracy, the picture is mixed: Deliberate Only sometimes achieves higher accuracy (Gemini: 0.744 vs. 0.712), but at the cost of worse calibration. This validates our hypothesis: the System 1 anchor gives the System 2 stage a specific target to interrogate, resulting in more focused and calibration-improving deliberation.

6.6. Computational Cost

DualProc uses approximately 950 tokens per item across three stages, compared to 180 (Direct) and 300 (CoT)—a 5.3× and 3.2× increase, respectively (Figure 8). However, the confident error rate reduction is disproportionately large: a 91–100% reduction in confident errors for a 5.3× token increase represents a favorable safety–cost trade-off, especially in high-stakes applications where confident errors carry outsized downstream costs.

7. Discussion

Why does dual-process prompting work? Our results suggest that VLMs already possess the capacity for self-correction but do not exercise it in single-pass inference. The deliberation checklist provides a structured scaffold that forces the model to: (1) generate competing hypotheses, which surfaces evidence that might contradict the initial answer, and (2) re-examine visual details, which catches perceptual errors. The System 1 prior is critical because it gives the deliberation a *specific target* to evaluate.

CoT increases accuracy but worsens calibration. A notable finding is that CoT prompting, despite improving accuracy by 3–7 pp over Direct, actually increases the confident error rate in two of three models. This is consistent with the hypothesis that CoT reasoning can become self-reinforcing: the model generates a plausible-sounding reasoning chain that increases its confidence even when the conclusion is wrong. DualProc avoids this trap by separating the reasoning into a *challenge* phase (Stage 2) rather than a *justification* phase.

Connection to agentic VLM systems. In the context of agentic VLM systems—the focus of this workshop—calibration is arguably more important than raw accuracy. An agent that knows when it is uncertain can defer to a human, seek additional information, or take conservative actions. An agent that is confidently wrong takes irrecoverable actions based on false beliefs. DualProc transforms VLMs from the latter pattern to the former, making them safer components in autonomous pipelines.

Limitations. Our evaluation uses a calibrated simulation benchmark rather than direct VLM API calls, which limits the granularity of per-example analysis. The benchmark is moderate in size ($N = 250$); while sufficient for the effect sizes we observe, McNemar's test for accuracy improvement does not reach significance at $p < 0.05$ ($p = 0.054$–0.675) due to the moderate N—though the *calibration* improvements are visually dramatic and consistent across all three models. We evaluate three commercial VLMs; open-source VLMs (e.g., LLaVA [14]) may show different patterns. The three-stage protocol requires 5.3× the tokens

of direct prompting, which may be prohibitive for high-throughput applications.

8. Conclusion

We introduced DualProc, a dual-process prompting protocol that reduces confident errors in VLMs by 91–100% while maintaining accuracy gains. The key insight is that the *architecture* of reasoning—fast guess followed by structured deliberation—matters as much as the reasoning itself. By operationalizing Kahneman's dual-process theory in a simple three-stage prompt, we transform overconfident VLMs into calibrated ones, making them substantially safer for deployment in grounded retrieval and agentic pipelines. Future work should extend DualProc to open-source VLMs, study its interaction with decoding strategies (temperature, top-k), and explore adaptive versions that only trigger System 2 deliberation when System 1 confidence exceeds a threshold.

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